



Scan for the package,
the do-file, and this poster

Coding Open-Ended Survey Text Without Leaving Stata: the catllm Package for LLM-Assisted Content Coding

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Abstract

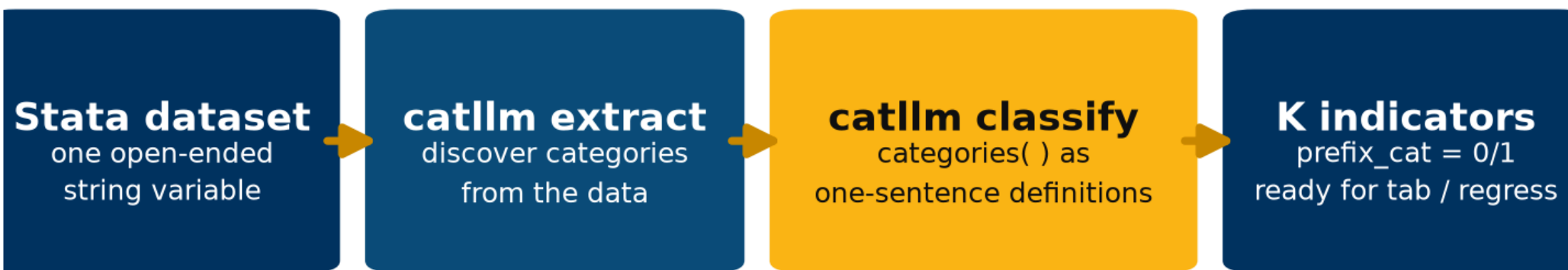
Open-ended survey items capture what closed-ended items cannot, and are routinely dropped from analysis because hand-coding them does not scale. `catllm` is a Stata command that codes free-text responses with large language models and returns ordinary Stata indicator variables. It wraps a Python backend (`cat-stack`) behind five verbs: `extract`/`explore` find categories, `classify` applies them, and `summarize`/`setup` close the loop, across OpenAI, Anthropic, Google, or Ollama. Tested on multi-label, multi-class items, proprietary models agree with human coders on **97%** of straightforward items and **88–91%** of complex interpretive ones; a multi-model ensemble reaches **98%** agreement with human consensus.

The Problem

- Researchers already use LLMs to code open-ended text, but little guidance exists for keeping classifications aligned with intended meaning without skewing conclusions.
- Model choice is often arbitrary. Researchers overpay for models that excel at general tasks, which rarely predicts coding accuracy.
- Coding several open-ended items at once means devising a bespoke strategy each time. That risks bias and error, not a documented method.
- Doing this from Stata isn't supported today: researchers export to CSV, script in Python, re-merge, and lose the audit trail that makes coding reproducible.
- Aim:** put the whole loop inside Stata, and tackle over-classification from several evidence-based angles: category definitions, model choice, and cross-provider ensemble consensus.

The Pipeline

Text goes in as a variable — categories come back as variables



Categories come from `extract` or your own list; `classify` returns multi-label indicator variables either way.

`generate()` is a prefix, not a variable name: a five-category scheme returns five 0/1 byte variables. `extract`'s specificity, iteration count, and category limit are tunable, so discovery matches the research question instead of a fixed default.

One Command

```
local cats ///
"Housing cost: Moved over rent, a mortgage, or affordability." ///
"Employment: Moved for a job, a transfer, or school." ///
"Family: Moved for caregiving or to be near relatives." ///
"Other: Does not fit any of the above categories."
```

```
catllm classify response, ///
categories(`cats') ///
apikey($OPENAI_API_KEY) ///
model("gpt-4o-mini") ///
generate(reason)
```

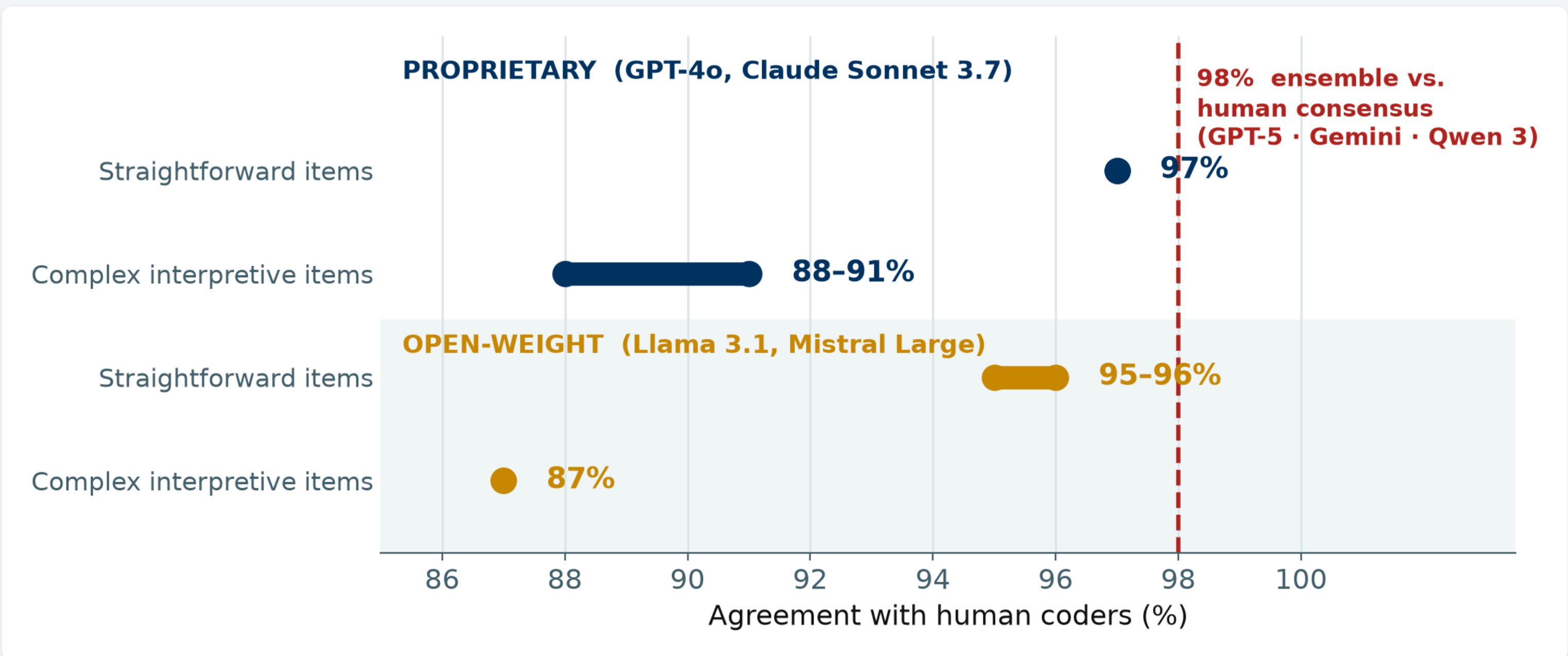
```
tab1 reason*
regress reason_Housing_cost age female
```

Category definitions, not one-word labels: writing each category as a sentence classifies several percentage points better on every model tested. An explicit `other` category is required, not optional — without it the model must guess on responses that genuinely fit nothing.

References

Soria (2026). *CatLLM: A Python package for generating, assigning, and scoring open-ended survey data and images*. | Soria (2026). *High agreement, different stories*. | Soria (2026). *Model diversity over model size*. | Bojic et al. (2025). *Comparing LLM and human annotators*. | Wei et al. (2023). *Chain-of-thought prompting*. | Dhuliawala et al. (2023). *Chain-of-verification*. | Zheng et al. (2024). *Step-back prompting*.

Accuracy Against Human Coders

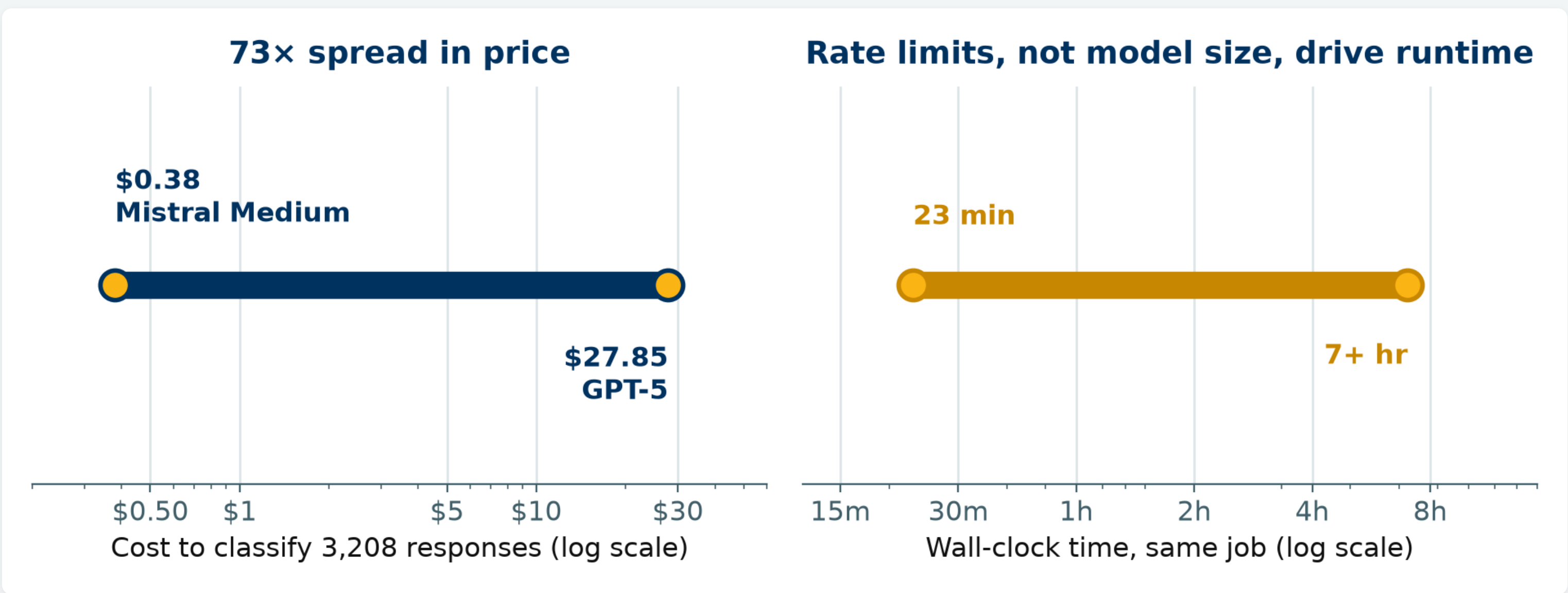


UC Berkeley Social Networks Study validation: four models against human annotators, across items of differing interpretive difficulty. Open-weight models trail proprietary ones by 1–2 points on straightforward items and converge with them on complex ones. Ranges span the models within each family.

What Moves Accuracy

- Task difficulty dominates model choice.** The gap between straightforward and interpretive items (≈ 9 points) is far larger than the gap between the best and worst model on a given item (1–2 points).
- Response brevity helps.** Responses under 50 characters are classified 7–11 points more accurately than long ones.
- Demographic variation is minimal.** Accuracy is stable across population segments — an essential precondition for using coded output in stratified analysis.
- Structured output is not the bottleneck.** Across 8 models $\times$ 3,208 responses (25,664 classifications), 100% of successful calls returned valid structured output; the 0–18 failures per model were transient server errors, not parsing failures.

Provider Comparison



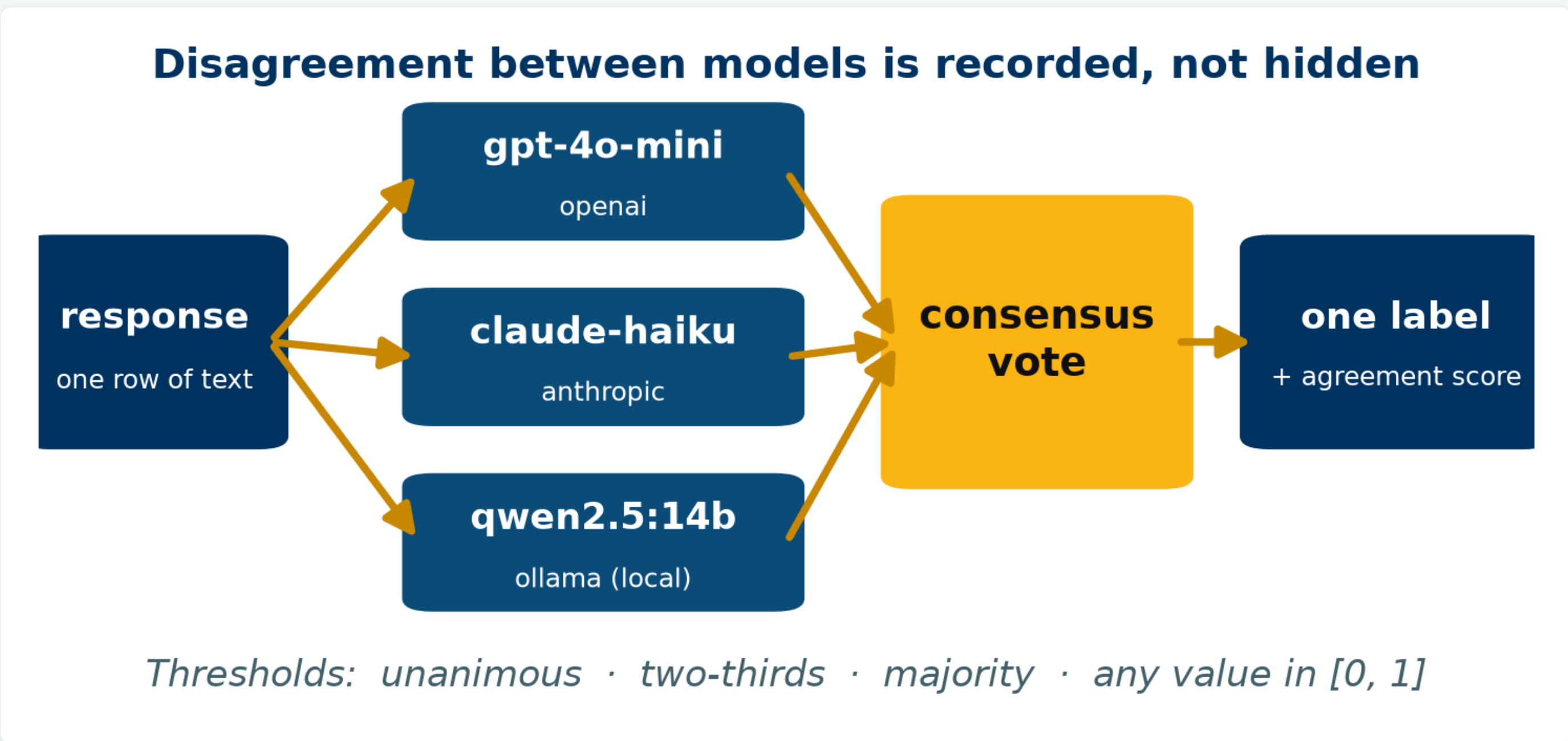
The same 3,208-response job, priced and timed across 8 models. Cost varies by a factor of 73 and wall-clock time by a factor of 18 — the latter driven by vendor rate limits rather than model size. Cost and accuracy are only loosely coupled, so the expensive model is frequently the wrong default.

Choosing a Route

Route	Cost	Speed	Data leaves machine
Frontier proprietary	High	Slow	Yes
Small proprietary	Low	Fast	Yes
Multi-vendor ensemble	Sum of parts	Slowest member	Yes
Local open-weight (Ollama)	\$0	Hardware-bound	No

For restricted-use data under an IRB or data-use agreement, the local route is often the only admissible one — and it is the route the package defaults to protecting, auto-starting the local server and enabling JSON recovery for smaller models.

Ensemble & Consensus Voting



`models()` takes several model/provider pairs; `consensus()` sets the bar for assigning a category. Cloud and local models can be mixed freely in one ensemble. Alongside each label the package returns an agreement score, so the rows where models split are identifiable rather than silently averaged away.

98%

ensemble vs.
human consensus

25,664

benchmark
classifications

\$0

fully local
route

Key Findings

• 97% / 88–91%

Agreement with human coders on straightforward vs. complex interpretive items. The ceiling is set by how interpretive the question is, not by which vendor you pay

• 98%

A three-model ensemble agrees with human consensus more often than any single model in it — disagreement between models is informative, not noise

• 73×

Cost spread across vendors for an identical job. Cost tracks accuracy only loosely, so the default choice should be justified rather than assumed

Reproduce This

Everything on this poster runs from one self-contained do-file — no data downloads, no file paths, guarded so it completes with a single API key or with none at all:

```
net install catllm, ///
from("https://raw.githubusercontent.com/chrissoria/cat-llm/main/stata-package/") ///
replace

catllm setup
do catllm_stata_conference.do
```

The file walks the full pipeline: classify, analyze the resulting indicators, compare three providers, run an ensemble with agreement scores, discover a coding scheme with `extract`, and repeat the whole thing locally at zero cost. Eight further worked examples ship with the package.

Limitations

- Concordance with human coders is not validity.** Where humans systematically misread a response, an agreeing model does too; agreement measures replication of a benchmark, not correctness.
- Validation is strongest on English-language survey text** of the kind studied here. Performance on other languages, registers, and domains needs separate checking on your own data.
- Models are moving targets.** A model version can be deprecated or silently updated, so the version string belongs in your methods section alongside the seed.
- Smaller local models are less tested.** The published benchmarks use high-end models; the low end warrants its own validation before use in production coding.